

SPEAKER IDENTIFICATION

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Group 22

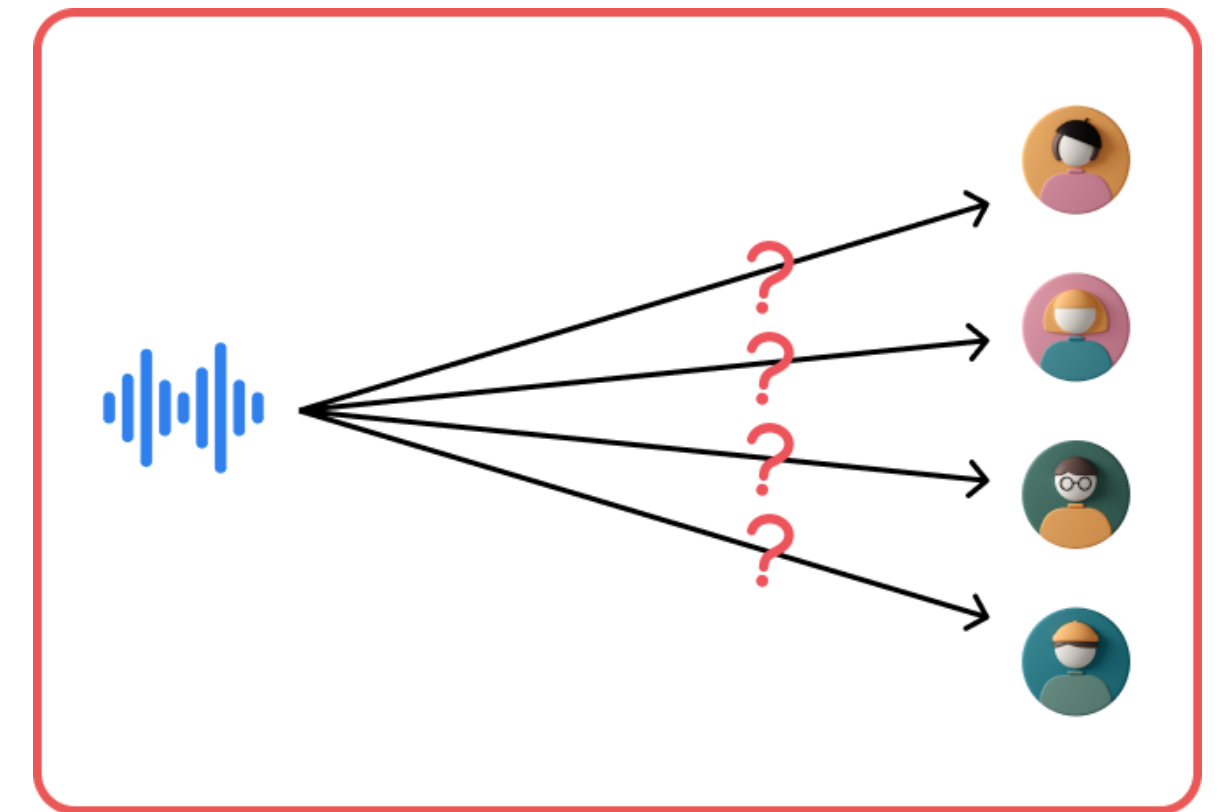
ML Project Presentation



Problem Statement

Speaker Identification using classical machine learning techniques

The advancement of voice recognition technology, especially in security, authentication, and virtual assistant applications, faces a substantial challenge in accurately identifying speakers within diverse acoustic environments. The complication is heightened by the presence of background noise, requiring a **dedicated exploration of techniques to improve speaker recognition in real-world situations.**

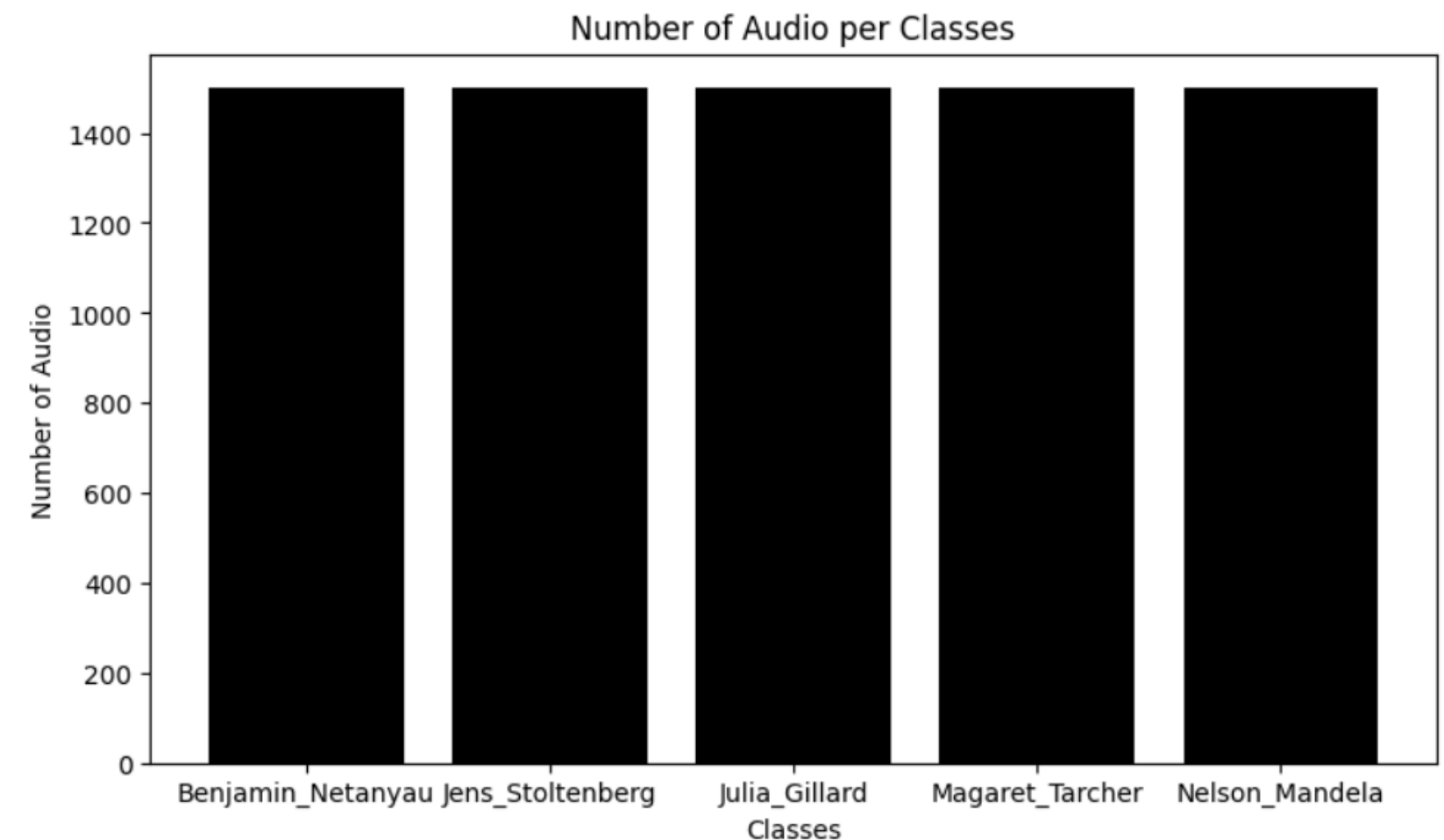


Dataset

Prominent Leader's Speeches

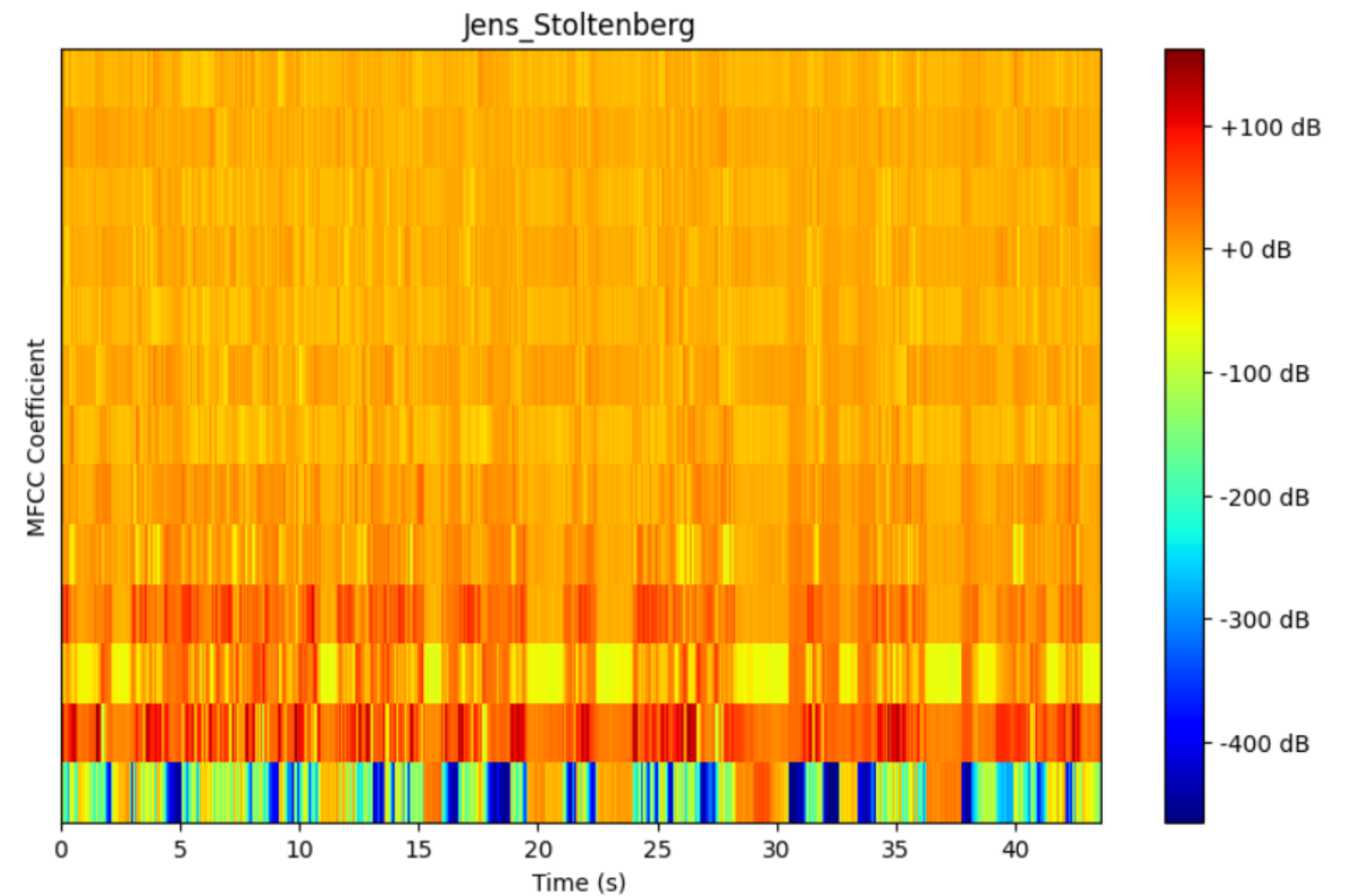
Description

- One-second audio clips extracted from speeches delivered by **five** distinguished country leaders—**Benjamin Netanyahu, Jens Stoltenberg, Julia Gillard, Margaret Thatcher, and Nelson Mandela**.
- These clips, encoded in the PCM format with a sampling rate of **16kHz**.
- The dataset has a separate folder containing audio files representing **background noise**, such as laughter and applause.
- The clips have **consistent sampling rate and PCM encoding**.



Exploratory Data Analysis

- The data is organised into a dataframe, **rows shuffled** for **randomness** to **avoid order bias**.
- **Feature extraction**- include Mel-frequency cepstral coefficients (MFCC), delta, double delta, and root mean square (RMS).
- **Data Augmentation**- Added background noise to the audio to make the dataset robust.
- Feature extraction from augmented data.



Methodology

01

Training - Gaussian Mixture Model

Our code employs Gaussian Mixture Models (GMMs) for speaker recognition, training a GMM for each label using the `train_gmm_and_save` function.

02

Testing

Testing has been done by listing audio files in a test folder, and the model predicts speaker labels using the trained GMMs, creating a mapping in a DataFrame.

03

Evaluation

The evaluation compares predicted labels with ground truth, calculating metrics like accuracy, classification report, and confusion matrix for model performance assessment.

Results

Our model uses Gaussian Mixture Models (GMM), achieving an impressive 99% accuracy. Precision, recall, and F1-score metrics for individual classes demonstrate the model's robust performance. The utilization of GMM aligns with established practices, emphasizing the efficacy of traditional machine learning techniques for superior speaker identification. This work contributes valuable insights for advancing speaker recognition systems.

	precision	recall	f1-score	support
0	0.99	1.00	1.00	301
1	0.99	0.96	0.97	300
2	1.00	1.00	1.00	300
3	0.96	0.98	0.97	300
4	1.00	1.00	1.00	300
accuracy			0.99	1501
macro avg	0.99	0.99	0.99	1501
weighted avg	0.99	0.99	0.99	1501

Comparative Analysis

1. While many existing models rely on deep learning (DL) techniques, our model demonstrates substantial improvement by exclusively leveraging traditional machine learning methods, emphasizing the pivotal role of feature extraction in achieving enhanced results.
2. In line with prevalent practices, our model incorporates Gaussian Mixture Models (GMM) for speaker identification, aligning with established approaches in the field. This integration showcases our commitment to adopting proven methodologies while showcasing advancements through innovative feature extraction techniques.

Conclusion

We conclude that **our model is upto the expected standards** and with this we conclude this phase. Our concise and robust methodology sets the stage for advancing speaker recognition in practical applications like security and authentication.

Our project **contributes to the evolution of speaker recognition technologies**, employing innovative approaches to overcome challenges. The "Prominent Leader's Speeches" dataset, enhanced with intentional background noise, mirrors real-world complexities.

Our concise and robust methodology paves the way for advancements in practical applications such as security and authentication.

Thank You

Learnings

Adapting - In the midst of the mid-term, a team member withdrew from the course. To address this, we responded by redistributing the workload among ourselves.

Improving - The Phase 1 demo highlighted the need for us to focus on the problem rather than fixating on the solution. We had been overcomplicating things, but the solution turned out to be simple and straightforward.

Contribution

Aadya- EDA, Report, Model

Paridhi- Existing Analysis, Report, presentation

Prakhar- EDA, Model

Shazra- EDA, Model